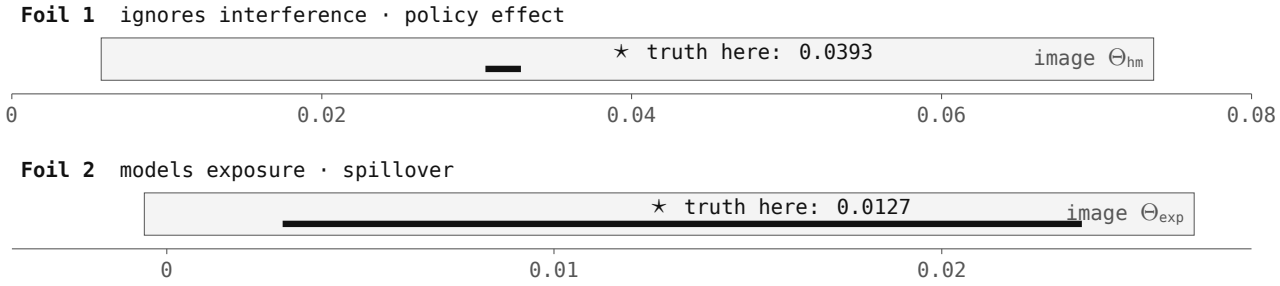
Not-ruled-out images of policy contrasts

A village agent-based model, calibrated by history matching, and two analysts (a field experimenter and a calibrator) graded against what the targets cannot rule out

The paper in one picture



The diagram shows the min-I point, the calibrator's best-fitting NROY point. The two rows use different scales. Bands: the image of the effect over 400 sampled NROY points. Black lines: the analyst's nominal 95% interval from one panel. Stars: the simulator's true value at this point. Foil 1 is tight, sits inside the band, and misses the truth. Foil 2 contains the truth, yet it also sits inside its band: narrower than what the targets leave open.

The question

Many parameter points fit the same 4 targets (the status quo plus one randomized pilot), and they imply different effects of seeding the hubs. What must a calibrated simulator report? How far can an interval computed at one point be trusted?

The answer

Foil 1 misses the truth at all three points, the real village included, and is F-OVERCONF at all three. Foil 2 contains the truth at all three, and is F-OVERCONF at both calibrated points. Report the image, not one point.

In numbers

Points evaluated in three history-matching waves / points kept (NROY)	3,000 / 1038
Image of the policy effect over the sampled NROY points	[0.0058, 0.0737]
Policy effect at the planted point c*	0.0383
Foil 1 estimate below the truth, at $c^\star$ / min-I / median-I	11.9% / 19.4% / 15.8%
Foil 2 interval coverage when the exposure map holds exactly	93.4%

Augusto González-Bonorino
Arizona State University
Applied project for ECN 593

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github.com/Bonorinoa/ecn593-applied-project
Deterministic · 55 tests · one command replicates it

A calibrated simulator gives a policy effect for each parameter point that fits the targets. Many points fit the targets equally well, yet they imply different effects. This note therefore reports the *image* of the policy effect over the sampled NROY points, not one number. The image follows the *reporting logic* of a partially identified set (Manski 2003, Tamer 2010), though it is not an identified set in their sense (Sections 11 and 13).

The targets are what a researcher could observe: the village in the status quo, plus one randomized pilot. The policy we want to predict, seeding the highest-degree people, is never a target. We then ask two analysts for an interval. The *field experimenter* has one panel from the real village, and her test is whether her interval is right about her own world. The *calibrator* fits the model and works at one best-fitting point, and her tests are whether her interval is right about the real village and how it sits against the image. Because the targets are planted, the true effect is known, and the test can grade both analysts.

Headline.

> Foil 1 misses the truth at all three points, the real village included, and is F-OVERCONF at all three. Foil 2 contains the truth at all three points, and is F-OVERCONF at both calibrated points; at the real village it has no verdict.

The analyst who ignores interference reports a tight interval that sits inside the image and misses the truth, even before any calibration. The analyst who models exposure contains the truth everywhere, yet at both calibrated points her interval is strictly narrower than the set of spillovers that the targets leave open.

The model is a village of 800 people on a fixed random graph, and each person has a preventive-behavior level in $(0, 1)$. A campaign seeds 10% of the people. Section 10 gives the numbers.

1 The planner's problem

A planner compares assignment rules (policies) that act on people who influence each other. Under interference, the outcome of person i depends on the whole assignment vector, so the effect of treating person i alone is not defined. The natural estimand is a contrast of rules,

$$\theta(\pi, \pi'; c) \;=\; \mathbb{E}_c\big[\bar{Y}_T(\pi)\big] \;-\; \mathbb{E}_c\big[\bar{Y}_T(\pi')\big]\,. \tag{1}$$

Here c indexes a complete data-generating process (a kernel), and the expectation is over the simulation noise that remains once c is fixed.

The planner does not know c. She has a class of kernels $\{f(\cdot; c) : c \in C\}$ and a short vector of moments z that she wants to match. History matching then *rules out* parameter points; it does not give a posterior over c. The decision-relevant object is not the kept set in parameter space but the image of θ over that set.

Two sources of doubt sit side by side, and we keep them apart. The first is calibration: which parameter points remain. The second is the analyst: a default estimator fits one

panel at one parameter point. For each analyst we report an interval and compare it with the truth and with its own image.

2 Objects

World, kernel and panel. The world is the graph A plus the covariates X. It is drawn once, and every simulation shares it. The kernel f maps a parameter point c, simulation noise ω and an assignment rule π to a panel (Y_T, D). The kernel is complete, in that it can run every arm on the same noise, and the incompleteness lives in c, not in unobserved counterfactuals inside f (Section 3). The code implements the kernel as vectorized algebra (`simulate_panel` in `kernel.py`).

One run of the kernel gives one panel,

$$\mathcal{D}(c, \omega, \pi) \;=\; \big\{(Y_{iT},\, D_i,\, X_i,\, A_{i\cdot})\big\}_{i=1}^{N}. \tag{2}$$

A parameter point is not a panel: one panel is one draw of ω. The kernel truth $\theta(c)$ uses $R = 200$ paired reps, whereas an analyst sees one panel.

Box. $C \subset \mathbb{R}^5$ is the a priori range of the free parameters $c = (\alpha, \beta, \gamma, \kappa, \sigma_\varepsilon)$. The vector η is fixed inside the kernel and is not in C.

Estimands. Let $\bar{Y}_T$ be the mean of Y_T over people. The code defines three estimands:

$$\theta_1(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_1] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0], \tag{3}$$

$$\theta_2(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_2] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0] \quad \text{(headline)}, \tag{4}$$

$$\psi(c) = \theta_2(c) - \theta_1(c) \quad \text{(the premium)}. \tag{5}$$

The arm π_0 seeds no one (control, the status quo), the arm π_1 seeds 80 people at random (random arm, the pilot), and the arm π_2 seeds the 80 people with the highest degree (degree arm, the policy to predict). The expectation is over simulation noise and over the random assignment in π_1. The code replaces each expectation with a mean over 200 paired reps, and we define θ on the terminal mean $\bar{Y}_T$, not on the path.

Two exposure contrasts, $\delta_{\mathsf{spill}}(c)$ and $\delta_{\mathsf{dir}}(c)$, are defined with the foils (Section 6.6). The letter τ is only the threshold, and the premium is ψ.

Moments and targets. $M : C \to \mathbb{R}^4$ is the map from a parameter point to 4 simulated summaries (Section 4). The targets $z \in \mathbb{R}^4$ are the values that $M(c)$ must match. With five free parameters and 4 moments, the model is underidentified on purpose: the question is which policy effects remain possible, not which c is true.

Implausibility.

$$I(c) \;=\; \big(z - \widehat{M}(c)\big)^{\top} V(c)^{-1} \big(z - \widehat{M}(c)\big), \qquad V(c) = \mathsf{diag}\big(\sigma_z^2 + \sigma_{\mathsf{MC}}^2(c) + \delta^{\circ 2}\big). \tag{6}$$

$I(c)$ is a variance-weighted sum of squared gaps, not a distance. The three terms in V are the Monte Carlo variance of the targets, the Monte Carlo variance of the candidate moment, and a model-error allowance, which we call the discrepancy δ.

NROY. In the code, NROY is a finite set: the union, over three Latin-hypercube waves, of the sampled points c with $I(c) \leq \tau$. It approximates the level set $\{c \in C : I(c) \leq \tau\}$ but does not enumerate it. Membership means we cannot rule out c. It does not mean that c equals the truth. NROY is not a posterior.

Images.

$$\Theta_{\mathsf{hm}} = \{\theta_2(c) : c \in \mathcal{S}\}, \qquad \Theta_{\mathsf{exp}} = \{\delta_{\mathsf{spill}}(c) : c \in \mathcal{S}\}, \tag{7}$$

where $\mathcal{S}$ is a declared uniform sample of 400 NROY points. The code reports each image as the range $[L, U]$ over $\mathcal{S}$: the declared empirical image. The range is not a distribution. Different NROY points can give different values of θ_2, but they need not.

Diagnostic points. There are three diagnostic points, and we write c_0 for any of them. The min-I point and the median-I point of NROY belong to the calibrator. Both depend on I only, never on θ. The third point is $c^\star$, the real village, and it belongs to the field experimenter (Section 8).

Foils. A foil is an estimator that an analyst fits on one panel at a diagnostic point c_0. Each foil is judged against the truth at c_0, against the truth at $c^\star$, and against its own image (Sections 7 and 8). A foil never sees NROY, so it is not a candidate for Θ_{hm}.

3 Potential outcomes, and what counts as data

Under SUTVA, $Y_i = Y_i(D_i)$ with $D_i \in \{0, 1\}$. One of the two numbers per person is missing. That missing number is the incompleteness in the data. A regression with robust standard errors quantifies within-point uncertainty under the analyst's stated assumptions. It is a ruler, not a villain.

This project drops SUTVA: peers influence each other through outcomes ($\gamma > 0$), so $Y_i = Y_i(\mathbf{d})$ with $\mathbf{d} \in \{0, 1\}^N$. The contrast $Y_i(1) - Y_i(0)$ is not defined unless the assignment of everyone else is named. The well-defined object is an overall policy contrast in the sense of Hudgens and Halloran (2008), namely $\theta(\pi, \pi'; c)$ from Section 1.

In the field, the other values of $\bar{Y}(\pi)$ are missing because there is one campaign. In the simulator, f is complete: run both arms on paired noise and subtract. What remains unknown is which $c \in C$ is right, so Θ_{hm} is a not-ruled-out image. It is not a Manski identified set, and it is not a set of possible $Y_i(0)$. It shares the *reporting logic* (do not replace a set by a convenient point) but not the theorem.

Only two objects have empirical analogs.

empirical analog	simulator object	who uses it
micro data of one campaign	one panel $\mathcal{D}(c, \omega, \pi)$	Foil 1, Foil 2
status-quo data and one pilot	z with its variances and δ	$I(c)$, NROY

The following objects are not data: the Latin-hypercube points, the Monte Carlo reps, the NROY points, the image sample and $\theta(c)$. They are probes of f, and NROY points are arguments of $\theta(\cdot)$, not extra surveys.

4 The targets: the status quo and one randomized pilot

The targets are what a researcher could observe. She sees the village as it is (the status quo, arm π_0), and she can run one randomized pilot (the random arm π_1). She never sees the degree arm π_2 beside either of them. That arm is the policy she wants to predict, so it is never a target. If it were, the targets would contain the answer to the planner's question.[1]

The code uses four moments, in a fixed order. Each is a mean over $R_m = 30$ paired reps, and rep r of the control arm and of the random arm share one state stream.

#	name	arm	definition
1	sq_level	control	mean over people of Y_T
2	sq_corr_x1	control	corr over people of (Y_T, x_1)
3	sq_peer_corr	control	corr over people of $\big(Y_{T,i}, (WY_T)_i\big)$, with $W = A/k_i$
4	pilot_level	random	mean over people of Y_T

Here x_1 is the first covariate and W is the kernel's row-normalized adjacency, the same peer mean that enters the update rule. The Monte Carlo variance of each moment is the sample variance over the reps (with one degree of freedom removed) divided by the number of finite values. The code drops non-finite correlations. If fewer than two values are finite, the variance is infinite and I rejects the point.

The targets are planted: $z := \widehat{M}(c^\star) = (0.2255, 0.0726, 0.0003, 0.2618)$, with 200 reps and its Monte Carlo variances. The file data/planted.json stores them. Planting supplies a known benchmark and makes a match plausible. It does not guarantee that a finite Latin-hypercube search keeps a point. It also makes the true effect known, and that is the only way to check an overlay.

The discrepancy δ and the threshold τ. The discrepancy is $\delta = (0.02, 0.05, 0.05, 0.02)$ in the order above: 0.02 for the two levels and 0.05 for the two correlations. It is a design tolerance, and it is not small next to the effect of interest: δ_1 and δ_4 are of the same

[1] The earlier design of this project did exactly that. Its targets used outcomes under both seeding policies, and their difference was the targeting premium. It is archived at the git tag v1-worldbank-moments and is not described further here.

order as $\theta_1(c^\star) = 0.03631$. The threshold is the 0.95 quantile of χ^2 with 4 degrees of freedom, one per moment, so $\tau = 9.4877$.

The paired-rep identity. Because the control and random arms share their state streams, `pilot_level` $-$ `sq_level` at $c^\star$ is $\theta_1(c^\star)$ (the two printed values are 0.03631 and 0.03631). This is intended: a randomized pilot identifies θ_1. The question is whether the model can extrapolate from the pilot to θ_2, the effect of seeding the hubs.

What the moments cannot do. The moments use two arms, not three, and the control arm appears only as the status quo. So θ_2, a contrast of the degree arm against control, is not a moment that the code matches. The code uses fewer reps for the moments than for the estimands because the search must be cheap, and the estimands are computed only for kept points. The targets and the estimands each use 200 reps. Equal replication counts do not imply equal Monte Carlo precision across these different statistics.

5 Order of operations

The order matters: if θ decides which c to keep, the image is a fit, not a report. For that reason the code fixes every exogenous parameter in `config/` before any wave runs.

1. Fix the kernel f, the box C, the targets z, the discrepancy δ, the threshold τ, the estimands and the rule for the diagnostic points.
2. Draw the world (A, X) once.
3. History matching. Evaluate $I(c)$ on a space-filling design and keep the points with $I \leq \tau$. Later waves search the box of the kept points; NROY is the union of the kept points. An empty NROY is a finding: the pair (f, z) is rejected. Never move τ or δ.
4. Leg A. Sample 400 NROY points, compute $\theta_2(c)$ and $\delta_{\text{spill}}(c)$ at each, and report Θ_{hm} and Θ_{exp} as ranges. Then evaluate the endpoint candidates again with fresh reps.
5. Leg B. At each of the three diagnostic points, fit the two foils on one panel and compare each interval with the truth at that point, with the truth at $c^\star$, and with its own image.

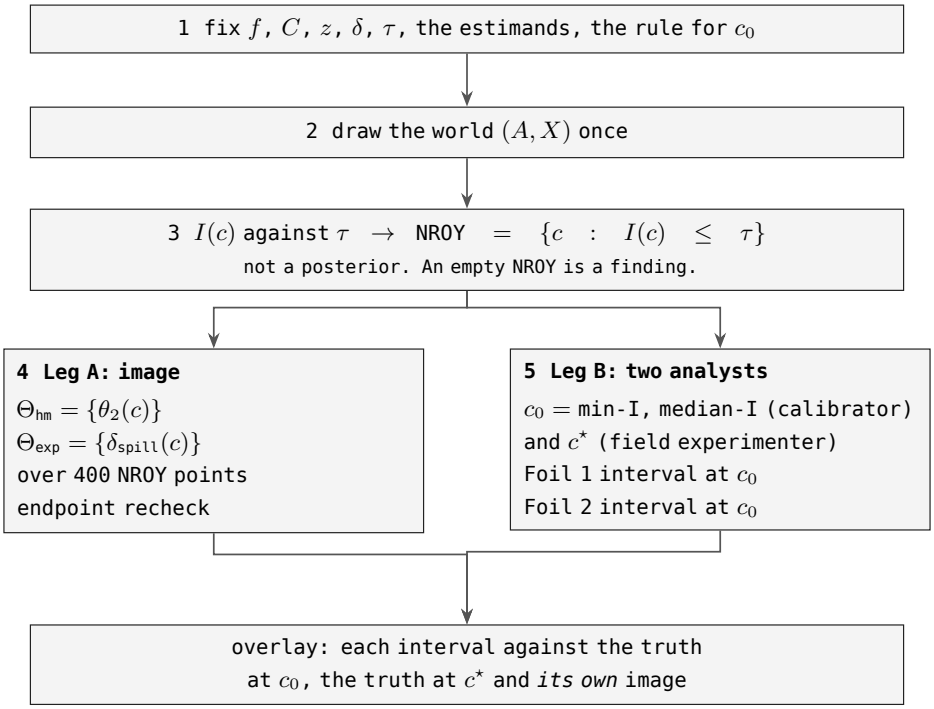


Figure 1: Order of operations. Legs A and B are two reports and do not merge into one object.

6 Design of exogenous quantities

The code fixes the history-matching quantities below before the waves run, whereas the later Foil 1 rescaling is post hoc and is disclosed in the results. The targets and δ are in Section 4.

6.1 Threshold τ. The code uses the 0.95 quantile of χ^2 with 4 degrees of freedom, $\tau = 9.4877$, as its design threshold. This choice does not assert that the implemented $I(c)$ has an exact χ^2 sampling law or define a credible region. It is a rule for ruling out. If we move τ after we see NROY, a filter becomes a fit. The same level, 95%, sets both foil intervals.

6.2 Discrepancy δ. The code adds the fixed discrepancy δ to V. In this planted exercise it is a design tolerance, not an estimated model-error variance. With external targets, the same slot is the allowance for a model that is not the world. If we loosen δ after a wave, it is the same error as moving τ.

6.3 The box C. The box is wide enough for underidentification ($5 > 4$) but also narrow: persistence, peer weight and the effect of seeding are positive, and the intercept does not push Y to saturation for every c. The value $\gamma = 0$ is outside the box on purpose, since the project is about interference. A world without interference that survives calibration is a separate finding.

6.4 How c is drawn. No one picks a favorite parameter point by looking at θ. Wave 1 is a Latin hypercube of 1000 points in C, and the code keeps each point with $I(c) \le \tau$. Waves 2 and 3 draw 1000 new points in the bounding box of the points kept in the previous wave. The box widens by 20% of its span and is clipped to C. There is no emulator: the code simulates every point. NROY is the *union* of the kept points over all waves. The planted point $c^\star$ is the center of C, and we chose it without looking at θ.

6.5 Diagnostic points. The min-I point is the closest match in NROY. The median-I point is the point at position $\lfloor |\text{NROY}|/2 \rfloor$ in the order of I, and both depend on moment fit, not on the estimand. The third point is $c^\star$ itself. We use three points, not 400. Running an analyst at every kept point gives the analyst the calibration set, and default practice does not do that.

6.6 Foils.

Foil 1: the SUTVA analyst. Foil 1 uses the degree arm and one panel. It fits OLS of Y_T on $Z_i = (1, D_i, X_i^\top)$, which has $k = 5$ columns, and the coefficient on D is $\hat{\beta}$. The standard error is HC1:

$$\widehat{\text{Var}}_{\text{HC1}}(\hat{\beta}) \;=\; \frac{n}{n-5}\,(Z^\top Z)^{-1}\Big(\sum_{i=1}^{n} \hat{u}_i^2\, Z_i Z_i^\top\Big)\,(Z^\top Z)^{-1}. \tag{8}$$

The interval for $\hat{\beta}$ is $\hat{\beta} \pm 1.96\,\mathrm{SE}$. On the degree arm, D is "being a top-degree person." The SUTVA analyst turns $\hat{\beta}$ into a policy estimate for the whole village, $(k/N)\hat{\beta}$, with $k/N = 0.1$, and the interval scales the same way. We call this interval J_1, and we compare J_1 with Θ_{hm}. The scaling by a known positive constant keeps the analyst's own coverage statement, but it does not repair the analyst's assumptions.

Foil 2: Hájek on the random arm. Foil 2 uses the random arm and one panel. The random arm has a known lottery, so we can write exposure probabilities before we see Y. The exposure share is

$$k_i = \max\Big(\sum_j A_{ij},\, 1\Big), \qquad s_i = \frac{\sum_j A_{ij} D_j}{k_i}. \tag{9}$$

The share s_i counts treated neighbors, not neighbor outcomes, and an isolated person has $k_i = 1$ and $s_i = 0$. The code defines four strata:

$$\begin{aligned} S_1 &= \{i : D_i = 1\}, \\ S_{0a} &= \{i : D_i = 0,\ s_i = 0\}, \\ S_{0b} &= \{i : D_i = 0,\ 0 < s_i < 0.25\}, \\ S_{0c} &= \{i : D_i = 0,\ s_i \geq 0.25\}. \end{aligned}$$

In a panel, if any stratum has fewer than 20 people (a thin cell), the code merges S_{0b} into S_{0c} for that panel. The kernel truth uses the realized stratum means of the true Y, averaged over 200 random-arm reps:

$$\delta_{\mathsf{spill}}(c) = \overline{Y}_{S_{0c}} - \overline{Y}_{S_{0a}}, \qquad \delta_{\mathsf{dir}}(c) = \overline{Y}_{S_1} - \overline{Y}_{S_{0a}}. \tag{10}$$

Foil 2 replaces each stratum mean with a Hájek mean, which weights each person by the inverse of the exposure probability,

$$\widehat{\mu}_S \;=\; \frac{\sum_i \mathbf{1}\{i \in S\}\, Y_i / p_{i,S}}{\sum_i \mathbf{1}\{i \in S\} / p_{i,S}}. \tag{11}$$

The first-order probabilities $p_{i,S}$ are exact: under the random arm, the number of treated neighbors of an untreated person of degree d is hypergeometric. The variance is the estimator of Aronow and Samii (2017). Its joint exposure probabilities come from 10,000 simulated random-arm assignments, and Young's bound covers pairs that never appear together. The interval is the estimate ± 1.96 SE. We call it J_2, and we compare it with Θ_{exp}.

The exposure map uses neighbor treatment, whereas the kernel peer term uses neighbor outcomes and the effect travels more than one step. So the map is not exact for the kernel. This limits Foil 2, but it is not a reason to replace Foil 2 with a second SUTVA analyst. Foil 2 does not estimate θ_2: the degree arm has no lottery, so given A its assignment probability is 0 or 1, and inverse-probability weights are not defined.

7 Interval geometry

Intervals and images are subsets of $\mathbb{R}$. A verdict is an inclusion, not a slogan. Let $J = [lo, hi]$ be the interval of a foil at one diagnostic point, let $\Theta = [L, U]$ be the image of the *same* kind of functional over the 400 sampled NROY points, and write $\text{int}(\Theta) = (L, U)$. The function `classify_interval` gives:

$$\text{F-HONEST} \iff lo \leq L \text{ and } hi \geq U \iff \Theta \subseteq J, \tag{12}$$

$$\text{F-OVERCONF} \iff lo > L \text{ and } hi < U \iff J \subset \text{int}(\Theta). \tag{13}$$

Every other case has no verdict, including an interval that overlaps the image with neither inclusion and an interval with an end that is not finite. Each foil goes against its own image: J_1 against Θ_{hm}, and J_2 against Θ_{exp}.

A verdict is a statement about interval geometry. F-OVERCONF means the analyst's interval is narrower than the set of effects that the targets leave open, and it does not say that the interval fails to contain the truth. The code reports two separate indicators for containment at each point. The flag `covers_truth` says whether the interval contains the truth at the point where the analyst works, and `covers_planted_truth` says whether it contains the truth at $c^\star$. At $c^\star$ the two coincide. Both are descriptive indicators for one realized panel. They are not estimates of repeated-sampling coverage, and the repeated-assignment check (Section 10.7) tests Foil 2 apart from them.

8 The evaluation: two analysts, two layers

Both foils run at all three points, but the three points are not three copies of one test. They serve two analysts who know different things and who are graded on different questions.

The field experimenter. She has one panel from the real village, which is the point $c^\star$. She has not calibrated anything, and she has no image. Her test is whether her interval is right about its own world: does it contain the truth at $c^\star$? This is layer (a). It depends only on the identification conditions of her estimator.

- Foil 1 needs SUTVA, unconfoundedness given X, and a linear (additive, constant-effect) outcome model. The deterministic top-degree assignment does not make the second condition true, and interference breaks the first.
- Foil 2 needs a correct exposure mapping and a known randomization. The randomization is known here. The mapping is approximate (Section 6.6).

The calibrator. She fits the model to the targets and works at one best-fitting point: the min-I point or the median-I point. She is tested twice. First, is she right about the real village? This is layer (b), and the flag is `covers_planted_truth`. It adds a maintained assumption to layer (a), namely "my world is the world": the point where she works is taken to be the village. Second, how does her interval sit against the image

of the whole NROY set? That is the verdict of Section 7.

analyst	question	field	what has to hold
experimenter	right about its own world? (a)	covers_truth at $c^\star$	the estimator's identification conditions
calibrator	right about its own world? (a)	covers_truth	the same conditions
calibrator	right about the village? (b)	covers_planted_truth	(a), plus "my world is the world"
calibrator	against the image	foil*_vs_*	the maintained assumptions of the image

What the image maintains. The image is honest only relative to what it maintains: the kernel, the box C, the discrepancy δ and the targets (with τ). Change any of them and the image changes. There is also a sampling caveat. The sampled image is the range over 400 points drawn from a finite set of kept points, so it can understate the image over the full level set of I. The endpoint recheck (Section 10) addresses only the opposite error: bias that pushes an endpoint outward because the extremizer was selected from noisy estimates. It does nothing about the first. A verdict of F-OVERCONF survives an understated image, since a larger image would still contain J strictly inside. A verdict of F-HONEST would not.

All of these are descriptive containment indicators for one realized panel. The only coverage evidence in the project is the Foil 2 repeated-assignment check, and its points are $c^\star$, the min-I point and the median-I point of this run.

9 Illustration: a planted village

One village has $N = 800$ people. The graph has an edge rule rather than a mean degree: the code draws $A_{ij} \sim \text{Bernoulli}(p)$ with $p = 4/N$ for every ordered pair, takes $A \mid A^\top$ and sets the diagonal to zero. An edge therefore appears with probability $1-(1-p)^2 \approx 8/N$, and the mean degree is about 8. The theory gives 7.97, and the graph in this project measures 8.085. The graph is drawn once with a fixed seed and is not calibrated.

$Y_{i,t} \in (0,1)$ is the level of preventive behavior. The covariates are $X \sim U[0,1]^3$: education or literacy x_1, resources x_2 and baseline propensity x_3. A rule seeds $k = 80$ people from $t = 0$, and the three rules are control (π_0), random (π_1) and degree (π_2). Ties in degree go to the lower index. In the degree arm, 35 of the 36 people at degree 12 are seeded, and the arm is deterministic given the graph.

The kernel is a logit linear-in-means update: a linear-in-means index passed through a logistic link. For $t = 0, \ldots, T-1$ with $T = 20$ and $t^* = 0$,

$$Y_i(t+1) \;=\; \Lambda\Big(\alpha + \beta Y_i(t) + \gamma \frac{(AY(t))_i}{k_i} + \eta \cdot X_i + \kappa D_i \mathbf{1}\{t \geq t^*\} + \sigma_\varepsilon\, \varepsilon_i(t)\Big), \tag{14}$$

with $\Lambda(z) = 1/(1+e^{-z})$, $\varepsilon_i(t) \sim N(0,1)$ and $Y_i(0) \sim U[0,1]$. Peers enter through the lagged outcomes of neighbors, not through neighbor treatment. The update rule is chosen, not

derived, and its lineage has three parts: Manski (1993) gives the linear-in-means form, Bramoullé, Djebbari and Fortin (2009) give the row-normalized peer mean, and Brock and Durlauf (2001) give the logistic link.

	box / value	role
α	$[-2.5, -0.5]$	intercept of the index
β	$[0.05, 0.60]$	own lag
γ	$[0.05, 0.80]$	weight on the peer mean
κ	$[0.25, 2.50]$	effect of being seeded (the same on every arm)
σ_ε	$[0.01, 0.40]$	standard deviation of the shock
η	$(0.05, 0.10, 0.02)$	covariate weights, fixed, not calibrated

The planted point is the center of C: $c^\star = (-1.5,\ 0.325,\ 0.425,\ 1.375,\ 0.205)$. The targets are $z = \widehat{M}(c^\star) = (0.2255,\ 0.0726,\ 0.0003,\ 0.2618)$ for sq_level, sq_corr_x1, sq_peer_corr and pilot_level. The discrepancy is $\delta = (0.02, 0.05, 0.05, 0.02)$, and the threshold is $\tau = 9.4877$.

Every random stream is seeded. Rep r of every arm shares one state stream (the start $Y(0)$ and all shocks) and has its own assignment stream, so the reps are paired and contrasts between arms have low noise. The whole pipeline is deterministic, and a rerun reproduces every number.

9.1 Three properties of the model.

The update map is a contraction. The logistic function has slope at most $1/4$, and the peer mean $(AY)_i/k_i$ is an average, so it is at most $\|Y\|_\infty$. For two states under the same assignment and the same shocks, one step moves the sup-norm distance by a factor of at most $(\beta+\gamma)/4$. Over the whole box, $(\beta+\gamma)/4 \le (0.60+0.80)/4$, which is below one. So for a fixed assignment the process has a unique stationary regime, and it forgets $Y(0)$ at a geometric rate: after $T = 20$ steps the starting state matters by at most that bound to the power T, far less than the smallest Monte Carlo SE in Table 1 (0.000006). There are no tipping points and no multiple equilibria, so the model is complete in the sense of Section 11. The kernel, the box and the horizon are all fixed before any run, and the bound uses only them.

Peers enter as an average. The peer term is a linear-in-means average of neighbors' lagged outcomes, the form of Manski (1993) on a network (Bramoullé, Djebbari and Fortin 2009). The average matters for the contraction. With a sum in its place, the peer slope would grow with a person's degree, and the bound above would no longer be below one for every point in the box. The mean degree is 8.085.

The graph is Erdős–Rényi, and nearly tree-like. The graph has an edge rule with no clustering built in. Its global clustering coefficient is 0.0096 (83 triangles over 25,918 connected triples), against 0.0100 expected for a random graph of this density. So

two people with a common friend are rarely friends themselves. This is the structure in which linear-in-means peer effects are identified from network data (Bramoullé, Djebbari and Fortin 2009), but this paper does not try to identify γ from data. The remark is only that peers' peers are mostly not peers.

10 Results

All numbers in this section come from the committed results in `results/`. Every one is a macro from `numbers.tex`.

10.1 History matching kept a large NROY set. Against the targets $z = \widehat{M}(c^\star)$, three waves kept 181, 449 and 408 points of 1000 each, and the union is $|\text{NROY}| = 1038$ (Figure 2). The minimum I in the three waves is 0.0274, 0.0874 and 0.0141, and the median I is 51.01, 11.23 and 13.54. The threshold is $\tau = 9.4877$. The set did not collapse to a point.

The closest NROY point has $I = 0.0141$, but it does not recover $c^\star$: its parameters are $\alpha = -1.5956$, $\beta = 0.4709$, $\gamma = 0.6853$, $\kappa = 1.2496$ and $\sigma_\varepsilon = 0.2150$. Fit does not pick the truth. This point is the min-I point by the declared rule, not because it won on θ. The median-I point, by contrast, has $I = 3.3437$ and a larger κ (1.9202).

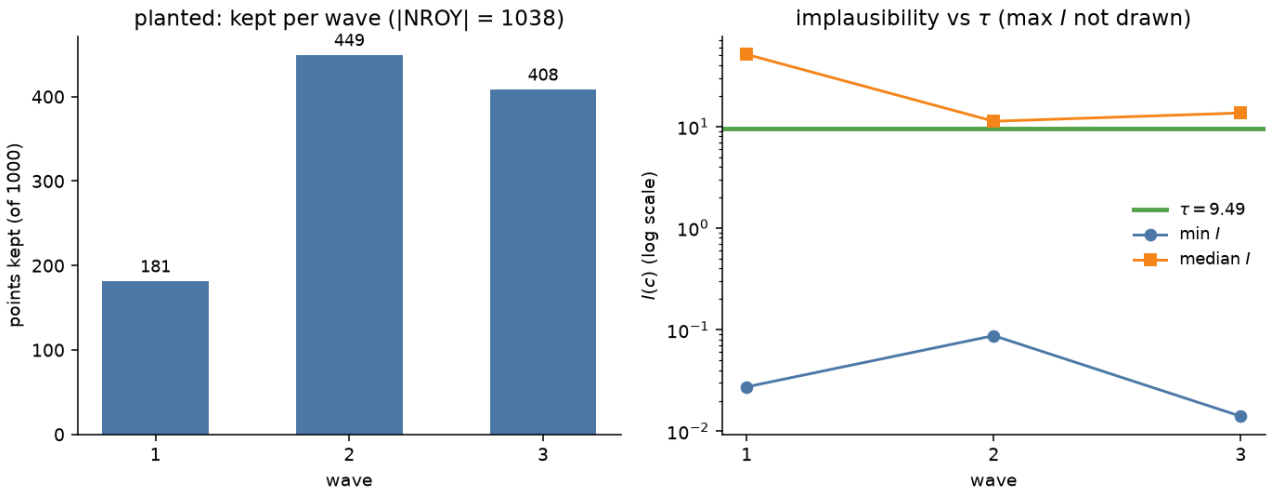


Figure 2: Planted history matching. Left: points kept in each wave. Right: min and median $I(c)$ against the threshold τ (log scale). Later waves concentrate on the kept region. The threshold does not move.

10.2 Which parameters the targets restrict. For each parameter, the code divides the standard deviation over the 400 sampled NROY points by the standard deviation of a uniform draw over C on that axis. A ratio near one means the moments do not restrict that parameter. This readout was declared before the run and is not a gate.

	α	β	γ	κ	σ_ε
spread ratio	0.259	0.965	0.992	0.962	0.839

Only α is pinned down. The ratios for β, γ and κ are close to one, and σ_ε is in between. The peer weight γ is the case the design most wanted to restrict (through `sq_peer_corr`), and its ratio is 0.992. That is below one only in the third decimal, so in practice the moments leave γ open. The pre-registration expected a ratio below one for γ, and expected the pilot moment to restrict the trade-off between α and κ more than the earlier design did. This paper does not rerun the earlier design, so it cannot check the second expectation against it. The sampled correlation between α and κ is -0.305, and the one between α and γ is -0.345. We report what the code found, and whether the expectation on γ counts as met is a call for the author.

10.3 The image of θ_2 is not a point. At the planted point, $\theta_1(c^\star) = 0.03631$ and $\theta_2(c^\star) = 0.03826$, and the premium is $\psi(c^\star) = 0.00195$. In this model, seeding the highest-degree people barely beats seeding at random, but that is a fact about this $c^\star$, not a theorem about targeting. The spillover at $c^\star$ is $\delta_{\text{spill}}(c^\star) = 0.00792$.

Over the 400 sampled NROY points, the headline image is $\Theta_{\text{hm}} = [0.00577, 0.07369]$, and the image of the spillover is $\Theta_{\text{exp}} = [-0.000578, 0.026525]$. The honest report for θ_2 is the range, not 0.03826. The range is not a sampling distribution and not a posterior.

Figure 3 shows every sampled point. Each bar is a pointwise Monte Carlo SE at a fixed c. For θ_2 the bars are smaller than the dots. For δ_{spill} they are visible.

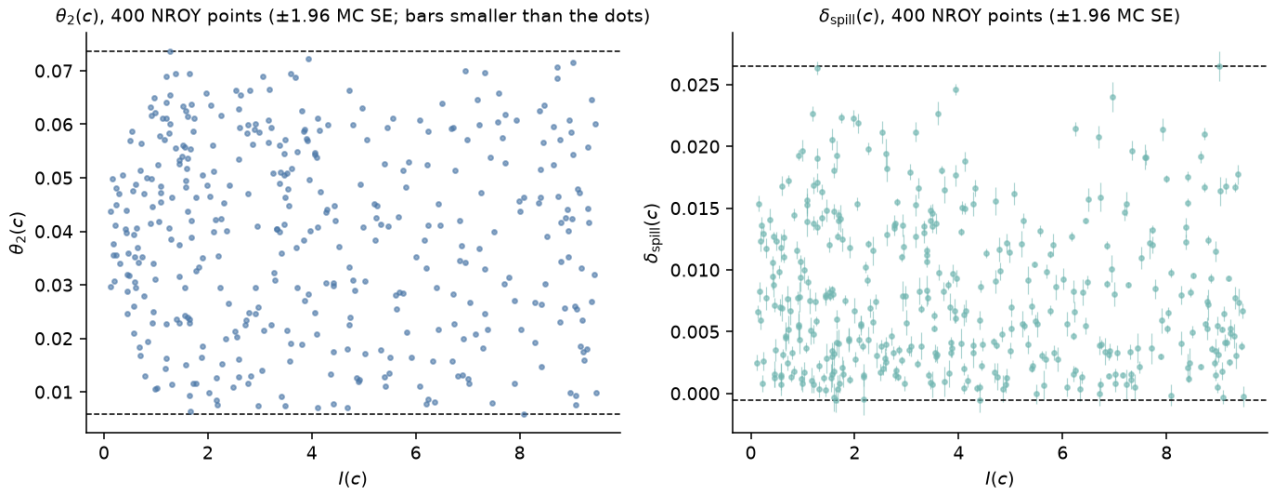


Figure 3: The 400 sampled NROY points against $I(c)$. Left: $\theta_2(c)$. Right: $\delta_{\text{spill}}(c)$. Bars show ± 1.96 pointwise MC SE. Dashed lines mark the image endpoints.

The endpoint recheck. The code evaluates again each point that sets an endpoint. It uses 200 fresh reps on separate streams. Table 1 gives the result.

Table 1: Independent recheck of the endpoint candidates (SE is the pointwise MC SE).

endpoint	NROY index	pilot (SE)	fresh (SE)
Θ_{hm} lower	293	$+0.005771$ (0.000008)	$+0.005772$ (0.000007)
Θ_{hm} upper	832	$+0.073692$ (0.000006)	$+0.073697$ (0.000006)
Θ_{exp} lower	987	-0.000578 (0.000505)	$+0.000200$ (0.000494)
Θ_{exp} upper	674	$+0.026525$ (0.000613)	$+0.027542$ (0.000601)

Each bar in image_points.json is a pointwise Monte Carlo SE conditional on a fixed parameter point c. The displayed image endpoints are extrema selected from 400 noisy estimates. So the SE attached to the extremizing point is not an SE for the endpoint. It does not include extremum-selection uncertainty. Fresh independent re-evaluation of the pilot-selected extremizers removes reuse-of-noise bias for those candidates. It is not simultaneous inference for the true extrema, and it says nothing about extrema that the sample missed.

10.4 The overlay: three points. Figure 4 puts each interval next to its own image at the three diagnostic points. The rows are the min-I point, the median-I point and $c^\star$, the village. The left column shows θ_2 against Θ_{hm} with the Foil 1 interval, and the right column shows δ_{spill} against Θ_{exp} with the Foil 2 interval. A star marks the truth at that point, and a triangle marks the truth at $c^\star$ (it is omitted in the $c^\star$ row, where it would sit under the star).

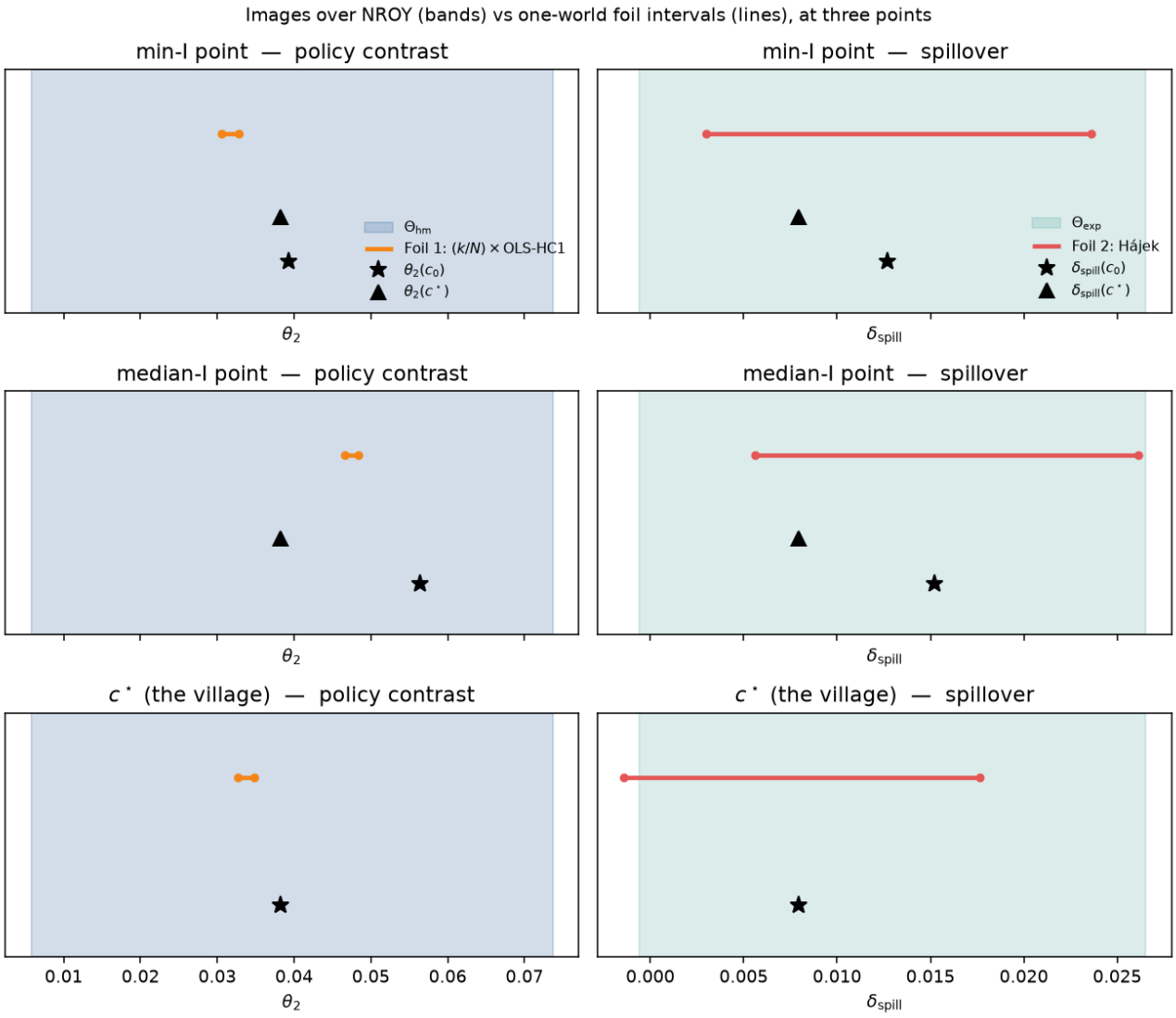


Figure 4: Images and one-world intervals at the three diagnostic points. Left: θ_2 against Θ_{hm} with Foil 1. Right: δ_{spill} against Θ_{exp} with Foil 2. Each interval is compared with its own image.

Table 2: The two foils at the three points. The first column is the field experimenter, and the other two are the calibrator.

	$c^\star$ (village)	min-I ($I = 0.0141$)	median-I ($I = 3.3437$)
truth $\theta_2(c_0)$	0.03826	0.03933	0.05642
Foil 1 interval J_1	$[0.03267, 0.03479]$	$[0.03057, 0.03285]$	$[0.04664, 0.04840]$
Foil 1 verdict	F-OVERCONF	F-OVERCONF	F-OVERCONF
contains truth at c_0 (a)	no	no	no
contains truth at $c^\star$ (b)	no	no	no
truth $\delta_{\text{spill}}(c_0)$	0.00792	0.01271	0.01520
Foil 2 estimate (SE)	0.00812 (0.00486)	0.01330 (0.00526)	0.01587 (0.00524)
Foil 2 interval J_2	$[-0.001405, 0.017638]$	$[0.002988, 0.023619]$	$[0.005598, 0.026135]$
Foil 2 verdict (pilot image)	no verdict	F-OVERCONF	F-OVERCONF
contains truth at c_0 (a)	yes	yes	yes
contains truth at $c^\star$ (b)	yes	yes	yes

10.5 What the two analysts find.

The field experimenter, at $c^\star$. Her Foil 1 interval is $[0.03267, 0.03479]$ and the truth is 0.03826, so the interval misses it, and the point estimate is 11.9% below the truth. Her interval is also strictly inside Θ_{hm}. The failure needs no calibration: it is layer (a), and it comes from the identification conditions of the estimator. Her Foil 2 interval $[-0.001405, 0.017638]$ contains the spillover 0.00792. It starts below the lower end of Θ_{exp}, so against the image it has no verdict.

The calibrator, at min-I and median-I. At both points the Foil 1 interval misses the truth at its own point (layer (a)) and the truth at the village (layer (b)), and it sits strictly inside the image. The Foil 2 interval contains the truth at its own point and the truth at the village, and it also sits strictly inside the image. At the median-I point the truth at the working point, $\theta_2 = 0.05642$, differs from the truth at the village, $\theta_2(c^\star) = 0.03826$, so the assumption "my world is the world" is costly there. At the min-I point the two are close (0.03933 and 0.03826), but the median-I point shows that membership in NROY does not guarantee it.

Containment and verdict are different tests. For Foil 2 they disagree. The interval contains the truth at every point, and yet at both calibrated points it is F-OVERCONF. There is no contradiction. An interval answers a question about one world, and the image answers a question about all the worlds that the targets cannot rule out. A one-world interval can be correct about its own world and narrower than the image at the same time. The calibrator who reports that interval as her uncertainty about the village would understate the uncertainty that the targets leave open. She happens to be right in this realization, and the image is what tells her she might not have been.

10.6 Foil 1: the SUTVA analyst. Table 2 shows that the Foil 1 interval is narrow. At $c^\star$, $\hat{\beta} = 0.3373$ with interval $[0.3267, 0.3479]$. At the min-I point, $\hat{\beta} = 0.3171$ with interval $[0.3057, 0.3285]$, and at the median-I point, $\hat{\beta} = 0.4752$ with interval $[0.4664, 0.4840]$. The scale factor is $k/N = 0.1$, and five statements stay with every report of Foil 1.

1. Foil 1 is not an identified estimator of the simulator's θ_2 under the actual interference model. It is the whole-village policy estimate implied by a SUTVA analyst, $(k/N)\hat{\beta}$, with $k/N = 0.1$.
2. That interpretation assumes consistency/SUTVA, conditional exchangeability and positivity given X, a correctly specified additive constant-effect outcome model, and a sampling/asymptotic basis for HC1. The deterministic top-degree assignment does not establish those assumptions.
3. After scaling, the Foil 1 intervals are strictly inside Θ_{hm} and exclude the truth at their own point at all three diagnostic points. So F-OVERCONF is an assumption-contingent interval-geometry finding. It is not a coverage claim for θ_2 under interference.
4. Across 200 paired panels, about 90% of the gap between θ_2 and (k/N) times the 200-panel mean OLS coefficient is the policy effect on unseeded people. For the displayed replication-0 point estimate, the corresponding shares are about 95% at $c^\star$, 93% at min-I and 92% at median-I.
5. The rescaling was adopted after the frozen numbers were inspected. Its justification is the SUTVA aggregation identity. The post hoc design correction is disclosed.

The gap decomposition. With paired streams, the policy effect splits exactly into a seeded part and an unseeded part. Let S be the seeded set, and let $y_2 - y_0$ be the effect on a person in one rep. Then

$$\theta_2 = \underbrace{\tfrac{1}{N}\textstyle\sum_{i\in S}(y_2 - y_0)}_{\text{the seeded part}} + \underbrace{\tfrac{1}{N}\textstyle\sum_{i\notin S}(y_2 - y_0)}_{\text{the unseeded part}}. \tag{15}$$

The unseeded part is zero under SUTVA. The seeded policy contrast is the mean of $y_2 - y_0$ over S, and under interference it is not a SUTVA effect of own treatment. Table 3 gives the numbers. The gap for the mean beta averages over 200 panels, whereas the gap for the displayed beta uses the one panel that Foil 1 sees. We keep the two apart.

Table 3: Gap decomposition for Foil 1 (200 paired reps).

	$c^\star$	min-I point	median-I point
θ_2	0.03826	0.03933	0.05642
seeded part	0.03396	0.03224	0.04824
unseeded part	0.00430	0.00709	0.00818
seeded policy contrast	0.3396	0.3224	0.4824
mean beta	0.3350	0.3146	0.4735
gap for the mean beta	0.00477	0.00787	0.00907
unseeded share of that gap	0.9028	0.9015	0.9015
OLS-error part of that gap	0.00046	0.00078	0.00089
gap for the displayed beta	0.00454	0.00762	0.00890
unseeded share of that gap	0.9488	0.9308	0.9185
baseline gap (seeded minus unseeded, Y under control)	0.0007	0.0007	0.0008

Most of the gap is the effect on people who were not seeded, and a SUTVA analyst has no term for it.

Why there is no forest foil. This was decision D3. In the degree arm, treatment is a deterministic function of degree and the index tie-break. Conditioning on degree leaves severe practical positivity failure: no treated people appear below degree 12, no untreated people appear above degree 12, and only one untreated person appears at degree 12. Omitting degree leaves the assignment variable unadjusted. A forest can differ numerically from OLS, but it retains the same SUTVA blind spot and does not identify θ_2 from this panel. A design-based third foil needs randomization across villages (Hudgens and Halloran 2008, Baird et al. 2018).

10.7 Foil 2: Hájek on the random arm. At each diagnostic point, `covers_truth` reports whether the displayed interval contains the simulator truth at that point, and `covers_planted_truth` whether it contains the truth at $c^\star$. Both are descriptive containment indicators for that realized panel, not estimates of repeated-sampling coverage and not validity gates. Foil 2 is checked separately by the pre-declared repeated-assignment calibration in `scripts/check_foil2_coverage.py`. All three Foil 2 intervals contain both truths (Table 2).

Against the saved 200-rep pilot image, $\Theta_{\text{exp}} = [-0.000578, 0.026525]$, the Foil 2 intervals at the min-I and median-I points, $[0.002988, 0.023619]$ and $[0.005598, 0.026135]$, are strictly interior and are classified F-OVERCONF. At $c^\star$ the interval reaches below the lower end of Θ_{exp}, so there is no verdict.

Does the endpoint noise change these classifications? The independent 200-rep recheck of the pilot-selected lower-end candidate is +0.000200 (pointwise MC SE 0.000494), against -0.000578 in the pilot. The recheck of the upper-end candidate is +0.027542 (SE 0.000601), against +0.026525. If we put the fresh values in place of the pilot endpoints, the lower end of each calibrated interval is still above the lower endpoint, the upper end is still below the upper endpoint, and the $c^\star$ interval still starts below the lower endpoint. So none of the three classifications changes. This is a comparison of printed

numbers, not a field of the results. The recheck evaluates selected candidates. It neither identifies nor bounds the true extrema over all 400 sampled NROY points, and it does not address an image that is too narrow.

An earlier version (commit `ebed7ce`) used a first-order variance that dropped the covariance between the strata, and commit `36ed3e6` replaced it with the Aronow–Samii variance.

The calibration check. The script `scripts/check_foil2_coverage.py` is a required check. Its gate uses outcomes that obey the exposure map exactly, with a true spillover of 0.03, and it draws 10,000 random-arm assignments. The gate rule was written before the run: the gate fails if the ratio of the mean estimated variance to the empirical variance, plus 1.645 bootstrap SE, is below 1. The gate passes, with a mean estimate of 0.03006 and a ratio of 1.034 (bootstrap SE 0.0154). The result:

> In a fixed-world repeated-assignment check whose outcomes obey the exposure map exactly, the nominal 95% Foil 2 Wald interval covered 93.4% of assignments (exact 95% binomial interval 92.9%–93.9%). The average estimated variance was 1.034 times the empirical design variance, so the shortfall is not explained by average variance underestimation. It reflects the finite-sample behavior of this studentized normal interval, with the implemented Hájek linearization and simulated joint exposure probabilities.

The check is specific to this implementation. It is not a test of the Aronow–Samii theorem. Their exact result is for Horvitz–Thompson totals with exact joint probabilities. The code uses Hájek means, a first-order linearization and one simulated table of joint probabilities.

The script also reports the kernel, where the exposure map is not exact because spillover also travels two steps. Over 200 kernel panels, the coverage of the nominal 95% interval is 0.920 at $c^\star$, 0.925 at the min-I point and 0.920 at the median-I point. If we redraw only the assignment and hold the noise fixed, the coverage is 0.930, 0.9267 and 0.9267. The code reports these numbers but does not gate on them. The gate result and these numbers are the only coverage numbers in the paper.

11 Relation to partial identification

This section is about the reporting logic only. Section 13 lists what the procedure is not.

Where the set comes from. In much of the partial-identification literature the set comes from an incomplete model. A game with multiple equilibria does not say which outcome occurs, so the model allows a set of outcomes, and the data bound the parameters without pinning them down (Tamer 2003, 2010). The set in this paper does not come from there. The kernel is complete: for each c and each draw of the noise it returns one panel, and Section 9.1 shows that the model has a unique stationary regime. The set comes from the

other source. There are too few moments for the parameters: 4 moments against five free parameters, and Section 10.2 shows that in fact only one parameter is restricted. Many complete worlds fit the same targets.

The image is a set under assumptions. The image is the set of effects under the maintained assumptions: the kernel, the box C, δ, the targets and τ. Adding one more assumption, $c = c_0$, collapses it to the single value $\theta_2(c_0)$. That is what the calibrator does when she works at one point. The trade is the one Manski (2003) calls the law of decreasing credibility: a stronger assumption gives a sharper answer and a less credible one. In Table 2 the sharper answer is the narrow interval at one point, and the credibility cost is the gap between $\theta_2(c_0)$ and $\theta_2(c^\star)$ at the median-I point.

What the targets restrict. How much of the set the targets remove is measured, not assumed. The spread ratios in Section 10.2 say which parameters the moments restrict.

12 Who will care, and who will not

Novelty at the level of slogans is weak. History matching already gives NROY in input space (Craig, Goldstein and Vernon [CITE?]). Equifinality of agent-based models is not new (Beven [CITE?]). Partial identification already says: report the set. Interference already has overall and direct effects (Halloran, Hudgens and Halloran [CITE?]). A regression with robust standard errors already gives an interval for a given panel. None of these pieces is new, so the remaining wedge is the composition: a named policy contrast under interference is reported as the image of a history-matched set, and two analysts are graded against it. The simulator is planted, so the truth is known. That is enough for an applied project, but it is not a methods paper.

Anyone who calibrates a simulator for policy. This matters if you publish a policy number from an agent-based model, a structural model or a microsimulation. Today you probably report the Monte Carlo error at one best parameter point. The overlay shows a split. Sampling error at one point is not uncertainty about which points the targets still allow. An empty sampled NROY is a usable negative result for the declared design: the run found no sampled point that passed the threshold for those targets. It does not establish infeasibility over all of C. You must not loosen the threshold. The width of the image is measured, not assumed. Here $\Theta_{\mathsf{hm}} = [0.00577, 0.07369]$.

Empirical causal inference. This is not a new identification strategy: it does not replace unconfoundedness, an instrument, difference-in-differences or a regression discontinuity, and it does not estimate effects that vary by covariate on a network. It does, however, matter in two narrow cases.

(1) You use a model as the identification vehicle, and you quote one $\hat{\theta}$. The model can be structural, an epidemic simulator or an agent-based model of a campaign. The

object to report is then the declared empirical image of θ over the sampled NROY points. A one-world interval quantifies within-point uncertainty under the analyst's stated assumptions. It is not the image.

(2) Interference is in the model, and the individual effect is not defined. The well-posed object is a contrast of assignment rules. A SUTVA analyst on a targeting arm, where D is a function of the graph, measures something near "being a top-degree person." That is an identification remark, not an estimator contest.

This project, specifically. The overlay is checkable because the targets were planted. The moments are the kind a researcher could observe, a status quo and one pilot, but the numbers are the simulator's own, so this is not an empirical estimate of any real campaign, and we do not present it as one. It shows that the composition can be run, with declared verdicts, including F-EMPTY and no verdict. As a contribution to empirical causal inference on real networks, it is a toy with a clean dictionary of objects. It becomes more only if someone supplies targets that this kernel can match, and those targets need a stated target population and a transport model. The threshold τ must not move.

13 What the procedure is not

- Not a posterior over c, and not a distribution of treatment effects. Θ_{hm} comes from a thresholded discrepancy, in which sampling noise is folded into V and the kernel is taken as true.
- Not a Manski or Tamer identified set. This is a finite empirical image produced by thresholding simulated moment discrepancies for a fully specified kernel. The analogy is only the reporting rule: do not collapse a set-valued result to one convenient point.
- Not a confidence set. A thresholded criterion function looks like the level sets of Chernozhukov, Hong and Tamer (2007), and the image of θ_2 over such a set is a projection, the object of Imbens and Manski (2004) and Kaido, Molinari and Stoye (2019). Those papers give coverage guarantees under their assumptions. Nothing here claims one, and V contains the discrepancy δ, which is a tolerance and not a sampling variance.
- Not a guarantee that the image is wide: its width is measured, not assumed.
- Not a claim that the closest NROY point recovers $c^\star$, which it does not: fit does not pick the truth.
- Not a coverage claim for either foil: F-OVERCONF is about interval geometry under the analyst's assumptions, and the containment flags describe one realized panel.
- Not a finding that analysts are bad at networks. Each quantifies within-point uncertainty under its own stated assumptions. The overlay asks whether one parameter point is enough.

- Not a claim about the real world: the project has one planted point, one graph and one set of targets, and it claims no external validity.

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Implementation. Repo `ecn593-applied-project`, branch `main`. `scripts/replicate.py` reruns everything. Each result file records the commit that produced it (`git_commit`). $|\mathsf{NROY}| = 1038$, $\Theta_{\mathsf{hm}} = [0.00577, 0.07369]$, $\theta_2(c^\star) = 0.03826$. Foil 1 is OLS-HC1 scaled by k/N. Figures are redrawn from the saved results. `README.md` lists the commands to replicate every number. The design is frozen in `docs/preregistration-v2.md`.